# Run R184 Synthetic Sampling Research Brief

HELIX FORGE LAB / p04-full-006940 / RUN R184 / 2026-07-13

| ALPHA F1 | PRECISION | RECALL | BETA F1 |
|---|---|---|---|
| 0.873 | 0.881 | 0.865 | 0.846 |

## Decision context

This report records a bounded synthetic sampling evaluation on HF-Eval-S1/S2 v3.2. The 2,400-record population is synthetic. Alpha M-14 is retained as a research checkpoint, not as a deployment authorization.

## Engineering interpretation

Aggregate improvement remains meaningful only with the frozen scoring policy. The review therefore keeps slice-level calibration, dataset lineage, and the comparison with Beta B-07 attached to run R184.

## Lineage

Upstream evidence: p04-full-000061 (downloads/inbox/2026-07-13/run-r184/dataset-assurance/specifications/run-r184-dataset-assurance-evaluation-contract-v01.xml).

| Slice | Records | Precision | Recall | F1 |
|---|---|---|---|---|
| slice-01 | 168 | 0.870 | 0.848 | 0.859 |
| slice-02 | 201 | 0.887 | 0.861 | 0.874 |
| slice-03 | 236 | 0.869 | 0.874 | 0.871 |
| slice-04 | 224 | 0.886 | 0.850 | 0.868 |
| slice-05 | 189 | 0.868 | 0.863 | 0.865 |